Fiscal Year:	FY 2016	**Task Last Updated:**	FY 05/21/2018
PI Name:	Schreurs, Ann-Sofie Ph.D.		
Project Title:	DNA Damage and Repair in Simulated Spaceflight-induced Bone Loss		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	94035	**Congressional District:**	18
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NASA Postdoctoral Program
Start Date:	10/01/2013	**End Date:**	09/30/2016
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Administration, USRA	**Contact Phone:**	
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Globus, Ruth Ph.D. (MENTOR/ NASA Ames Research Center)		
Grant/Contract No.:	NASA Postdoctoral Program		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	NASA Postdoctoral Program Fellow The spaceflight environment poses multiple challenges to the human body, including microgravity and ionizing radiation, which in some respects resembles aging. Microgravity and radiation may lead to excess production of reactive oxygen species (ROS) and resulting oxidative stress and tissue damage. In bone, elevated ROS can contribute to excess bone resorption by osteoclasts over bone formation by osteoblasts, and ultimately, osteoporosis. Therefore, we plan to use multiple approaches to determine the role of ROS and resulting oxidative and DNA damage in spaceflight induced bone loss. (1) Mechanisms of spaceflight induced bone loss: We will use transgenic mice, which we have validated to express the antioxidant catalase to the mitochondria in bone and bone cells.		

	(2) Bone loss prevention: we plan to test various interventions that have been shown with previous data to be able to mitigate the early increase in pro-osteoclast signaling, oxidative stress, and inflammation to find countermeasure for bone loss.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	The spaceflight environment poses multiple challenges to the human body, including microgravity and ionizing radiation, which in some respects resembles aging. Microgravity and radiation may lead to excess production of reactive oxygen species (ROS) and resulting oxidative stress and tissue damage. In bone, elevated ROS can contribute to excess bone resorption by osteoclasts over bone formation by osteoblasts, and ultimately, osteoporosis. Therefore, we used multiple approaches to determine the role of ROS and resulting oxidative and DNA damage in spaceflight induced bone loss: (1) Mechanisms of spaceflight induced bone loss: (a) We used a transgenic mice, which we have validated to express the antioxidant catalase to the mitochondria in bone and bone cells. Furthermore, we have shown that these transgenic mice, mCAT mice, had reduced oxidative damage in the bone compared to the wild type (WT) littermate after spaceflight simulation treatment. Finally we describe the genotype effects as well as the response of the WT and mCAT mice after spaceflight simulation in terms of skeletal microarchitecture. (b) In order to tease out the effects of each spaceflight simulation treatment, microgravity and radiation, we decided to look at oxidative damage, DNA damage, and cell cycle as well as pro-osteoclast gene expression. Our initial observations do show that indeed microgravity and radiation have differential responses and seem to use different pathways. (2) Bone loss prevention: (a) Based on our previous data, we decided to test various interventions that were able to mitigate the early increase in pro-osteoclast signaling, oxidative stress, and inflammation to find countermeasure for bone loss. We tested various antioxidants diets, antioxidant injections, anti-inflammatory injections, and dried plum diet. We have shown that the dried plum diet completely prevents radiation-induced bone loss, most probably by reducing the early pro-osteoclast cytokines signaling, and protecting the osteoprogenitors bone forming cells. (b) In vitro protection of bone forming osteoblasts from radiation. We tested various antioxidants to protect osteoblasts from radiation, and found that superoxide dismutase (SOD) protects osteoblasts from low Linear Energy Transfer (LET) such as gamma irradiation, but not from high-LET irradiation such as Iron. This provides a platform to look at the differences in terms of oxidative damage and DNA damage between low-LET and high-LET irradiation. In summary, all our data provides further insight in the mechanisms of spaceflight-induced bone loss and we found a potential novel radiomitigant.
Bibliography Type:	Description: (Last Updated: 02/05/2026)
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